

Sam Nguyen

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EDUCATION

Iowa State University

Bachelor of Science, Mechanical Engineering, 4.00 GPA - President's List 2x

Expected Graduation: May 2029

Ames, Iowa

WORK EXPERIENCE

Thingz

May 2026 – Aug. 2026

Mechanical Engineering Intern

Galena, Illinois

- Designed and managed large electrical substations in Autodesk Inventor by aligning and interpreting 30M+ point LiDAR scans to produce a digital twin representative of real-world equipment and environment
- Reduced design turnaround time by roughly 70% by rapidly building and organizing a standardized library of substation parts and assemblies, streamlining the CAD process and providing reference models for future projects
- Developed accurate, detailed models of electrical substation equipment, producing 1,000+ parts and sub-assemblies
- Operated 360° LiDAR scanners and cameras on-site to capture point cloud data across an entire 5+ acre substation
- Worked cross-functionally to coordinate modeling efforts and deliver weekly verbal and written progress reports
- Identified a productivity gap and resolved it by creating assembly identifier flow charts, diagrams, and documentation
- Consistently adhered to modeling principles and quality standards to ensure compatibility between new and existing parts, assemblies, and naming conventions

EXTRACURRICULAR

Formula SAE Electric Vehicle (FSAE EV)

Sep. 2025 – Present

Suspension Subsystem Member

Iowa State University

- Independently designed and manufactured a strain gauge calibration fixture, involving statics calculations, hands-on fabrication, and welding, culminating in significant data acquisition for the team's R&D of strain gauges
- Designed and additively manufactured jig tooling to ensure precise location of critical suspension geometry during production of components such as the A-arms, push rods, and tabs on all 4 corners of the vehicle.
- Oversee and own the bell crank/rocker assembly of the latest design cycle, using design tables and proper modeling practices to validate decisions alongside improving knowledge transfer and documentation
- Gained hands-on experience and certifications on a wide range of fabrication equipment across multiple projects
- Analyzed bearing specifications to determine the most efficient solution, balancing packaging and cost constraints
- Collaborated with external design team to ensure seamless integration between Blade Holder and Servo-to-Blade Adapter
- Modeled concept parts parametrically, enabling scalability and rapid design iteration to verify packaging concerns for Blade Holder and bearing integration

SolidWorks Club

Aug. 2025 – Present

Sophomore Representative

Iowa State University

- Achieved SolidWorks CSWP Certification with a flawless 100% through weekly CAD assignments and a rigorous 3-hour final assessment covering part modeling, design iteration, and assembly skills
- Evaluated engineering drawings of models and assemblies for compliance with ASME Y14.5 standards
- Elected Sophomore Representative after first year of membership, demonstrating rapid growth and mentorship
- Provided 1-on-1 instruction to club members on core SolidWorks skills and design best practices

PROJECTS

Fishing Pole Holder

Sep. 2025 – Nov. 2025

- Assessed design requirements using digital calipers and paper sketches to ensure proper product function
- Created and 3D-printed the final solution using SolidWorks and Bambu Lab X1C, material choice of PETG

SKILLS AND CERTIFICATIONS

SolidWorks CSWP, Autodesk Inventor, Additive Manufacturing (FDM 3D Printing), Python, Microsoft Office (Excel, Word, PowerPoint), Metal Lathe, MIG Welding, Grinder & Sander, Horizontal & Vertical Bandsaw, Table Saw, Mill, Drill Press

INTERESTS: Fishing, Sketching, Digital Cameras, Automotive, Astronomy, Soccer, Filmmaking, Time Management